

Lecture 12 - Advanced Topics

CPE241 – Database Systems

5 May 2026

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Term Project Presentation

- 12 May 2026, 13.30-17.30
- 8-10 minutes / group (not include Q&A)
- Scope: CRUD & Reports

Criteria	Excellent (5)	Good (4)	Acceptable (3)	Poor (1-2)
Requirement Summary	Clear and concise	Mostly clear	Some unclear points	Vague or missing
Interface & Query Demo	Effective, mapped to goals	Mostly functional	Limited demo	Unclear or absent
Optimization Discussion	Insightful and data-backed	Mentioned briefly	Minimal or unclear	None
Presentation Quality	Well-timed, engaging, visual	Minor flow issues	Read-heavy or rushed	Poor delivery

Term Project Report & Reflection

- Submission: **20 May 2026**
- Additional parts to complete your full report

Criteria	Excellent (5)	Good (4)	Acceptable (3)	Poor (1-2)
Interface + CRUD	Functional and well-mapped to DB	Mostly functional	Partial functionality	Incomplete or broken
Query Examples	Meaningful, efficient, varied	Mostly accurate	Basic queries only	Little or incorrect queries
Optimization	Indexed + justified + tested	Basic indexing used	Index mentioned only	No optimization
Clarity and Structure	Professional and organized	Minor formatting issues	Some disorganization	Unreadable or rushed

- Term project reflection form will be posted.
- Individual submission on 20 May 2026.

Today's Goals

- OLTP
- OLAP
- Distributed DB
- Cloud DB
- Data Warehouse
- Data Lake

OLTP

- Online **Transaction** Processing
- Examples
 - Banking systems (transferring money, checking balance, etc.)
 - Online shopping (placing an order, updating inventory, etc.)
 - Reservation systems (airline tickets, hotel booking, etc.)
- A transaction is a sequence of **one or more** SQL operations treated as a unit.
- Partial transactions are **not** allowed.

Move THB100 from A's account to B's account



Transaction:

- Check whether A has THB100.
- Deduct THB100 from A's account.
- Add THB100 to B's account.

OLTP

- Focus: managing real-time transactional data (inserting, updating, deleting records)
- Main features
 - Real-time transactions (minimal latency)
 - High volume, short transactions
 - Highly normalized RDBMS schemas (reduce redundancy)
 - Indexing data to improve response times
 - Concurrent users

OLTP

- Requirement: strong ACID properties

- **A**tomicity - all or nothing
(All actions in the transaction happen, or none happen.)
- **C**onsistency – it looks correct to me
(If each transaction is consistent and the DB starts consistent, then it ends up consistent.)
- **I**solation – as if alone
(Execution of one transaction is isolated from that of other transactions.)
- **D**urability – survive failures
(If a transaction commits, its effects persist.)

Case: Take THB100 out of A's account
but then there is a power failure before
the transfer to B's account

Q: What should be the correct state of
A's account
after both transactions' abort?

T1

BEGIN

$A = A - 100$

$B = B + 100$

COMMIT

T2

BEGIN

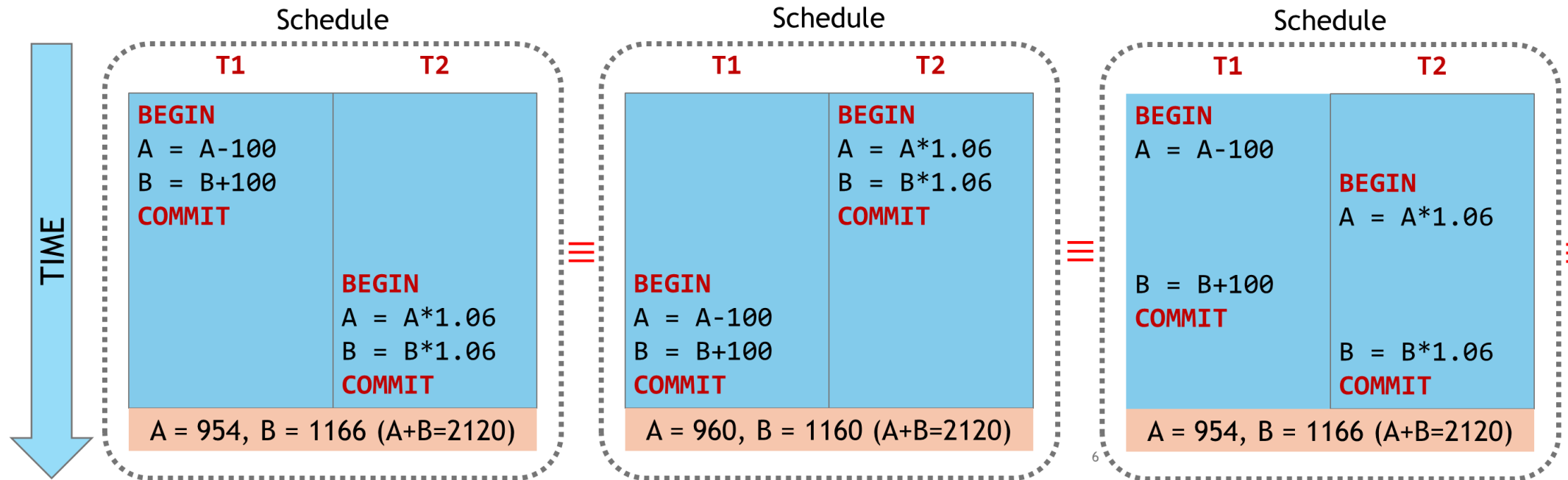
$A = A * 1.06$

$B = B * 1.06$

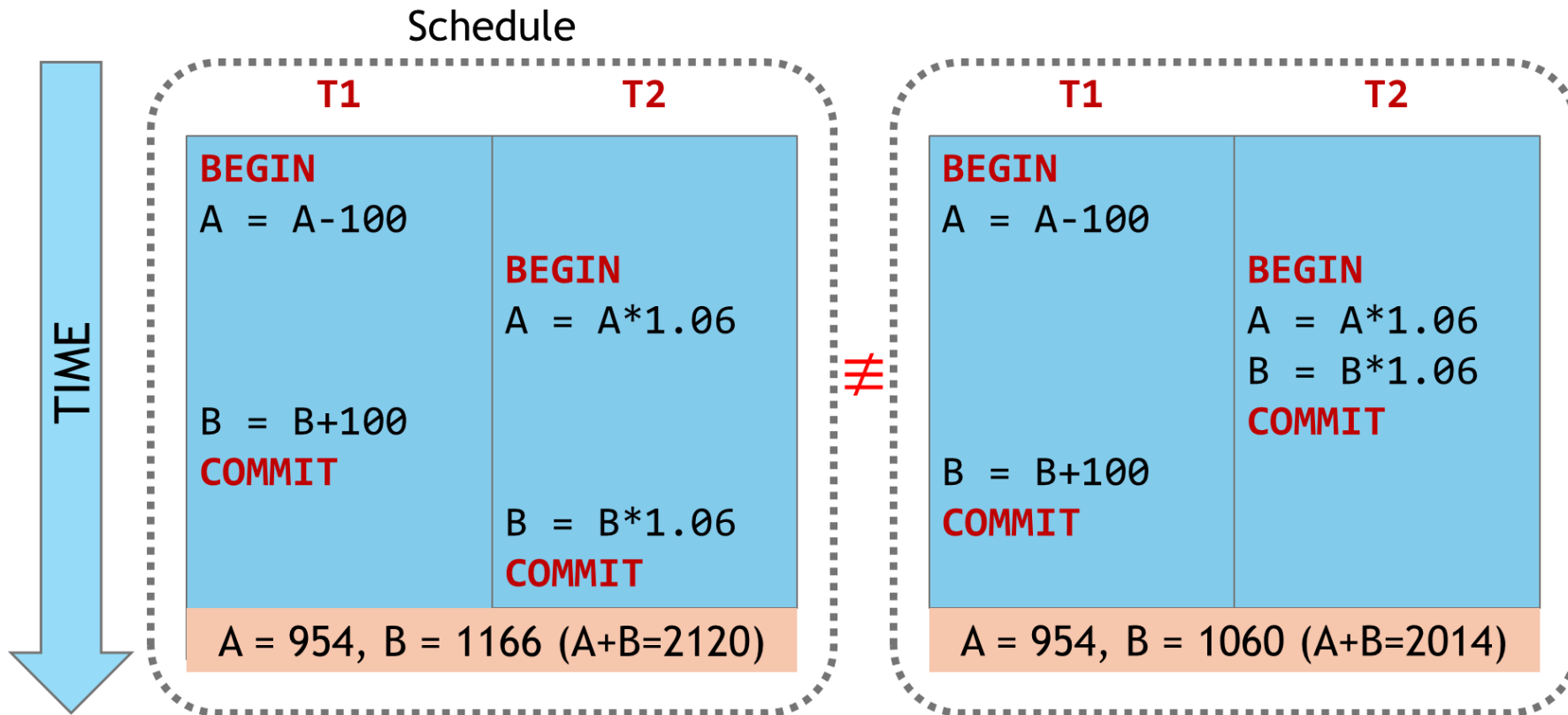
COMMIT

ACID - Isolation

- Example
Initial balance: A = 1000, B = 1000



ACID - Isolation – Bad Interleaving



OLAP

- Online **Analytical** Processing
- Examples
 - Executive dashboards
 - Sales reporting
 - Market research analysis

- Focus: analyzing large volumes of historical data to support decision-making.



Source: <https://techterms.com/definition/olap>

ONLINE ANALYTICAL PROCESSING (OLAP)

OLAP

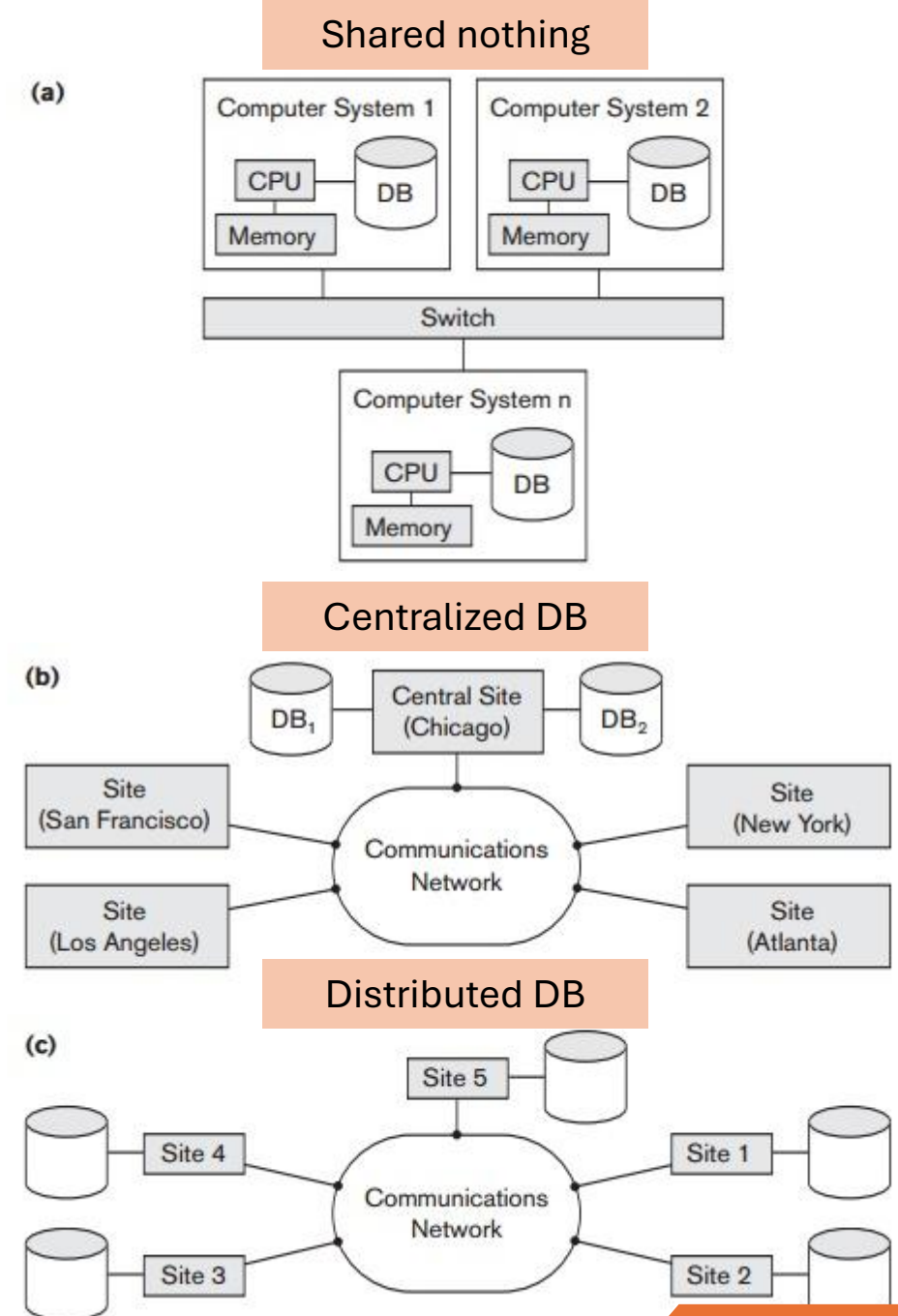
- Main features
 - Supports trend analysis, forecasting, and business intelligence (BI)
 - Complex, read-intensive queries
 - Aggregated, summarized data
 - Often organized into multidimensional cubes (e.g., time, location, product)
 - Optimized for rapid reading and aggregations
 - Denormalized schemas (Star Schema or Snowflake Schema)

OLTP & OLAP

Feature	OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing)
Purpose	Manage real-time daily transactions	Support complex data analysis and decision-making
Operations	Insert, Update, Delete, Simple Queries	Complex Queries, Aggregations, Data Mining
Data Freshness	Always up-to-date (real-time)	Historical or periodically updated data
Transaction Type	Short, frequent, atomic	Long, complex, read-heavy
Schema Design	Highly normalized	Denormalized (Star Schema, Snowflake Schema)
Data Volume	Relatively small per transaction	Very large (millions/billions of records)
Users	Customers, clerks, operational staff	Analysts, executives, managers
Performance Focus	Fast read/write, concurrency	Fast read and complex aggregation
Backup/Recovery Focus	Critical for each transaction	Less critical per query, batch backup often used
Example Systems	MySQL, PostgreSQL, Oracle (standard use)	Microsoft SSAS, Oracle OLAP, Snowflake

Distributed Database

- Think about your term project.
What if your system is for world-wide users?
- A collection of multiple, logically interrelated databases distributed across different physical locations but appearing to users as a single database.
- Types (considering data, data models, OS, DDBMS)
 - Homogeneous
 - Heterogeneous



Types of Distribution

- **Horizontal Partitioning (Sharding):**
Split data by **rows** across multiple nodes (e.g., user data partitioned by region)
- **Vertical Partitioning:**
Split data by **columns** (e.g., personal info on one server, transaction history on another).
- **Replication:**
Duplicate copies of data across nodes for reliability and performance.

Original Table

CUSTOMER ID	FIRST NAME	LAST NAME	FAVORITE COLOR
1	TAEKO	OHNUKI	BLUE
2	O.V.	WRIGHT	GREEN
3	SELDA	BAĞCAN	PURPLE
4	JIM	PEPPER	AUBERGINE

Vertical Partitions

VP1

CUSTOMER ID	FIRST NAME	LAST NAME
1	TAEKO	OHNUKI
2	O.V.	WRIGHT
3	SELDA	BAĞCAN
4	JIM	PEPPER

VP2

CUSTOMER ID	FAVORITE COLOR
1	BLUE
2	GREEN
3	PURPLE
4	AUBERGINE

Horizontal Partitions

HP1

CUSTOMER ID	FIRST NAME	LAST NAME	FAVORITE COLOR
1	TAEKO	OHNUKI	BLUE
2	O.V.	WRIGHT	GREEN

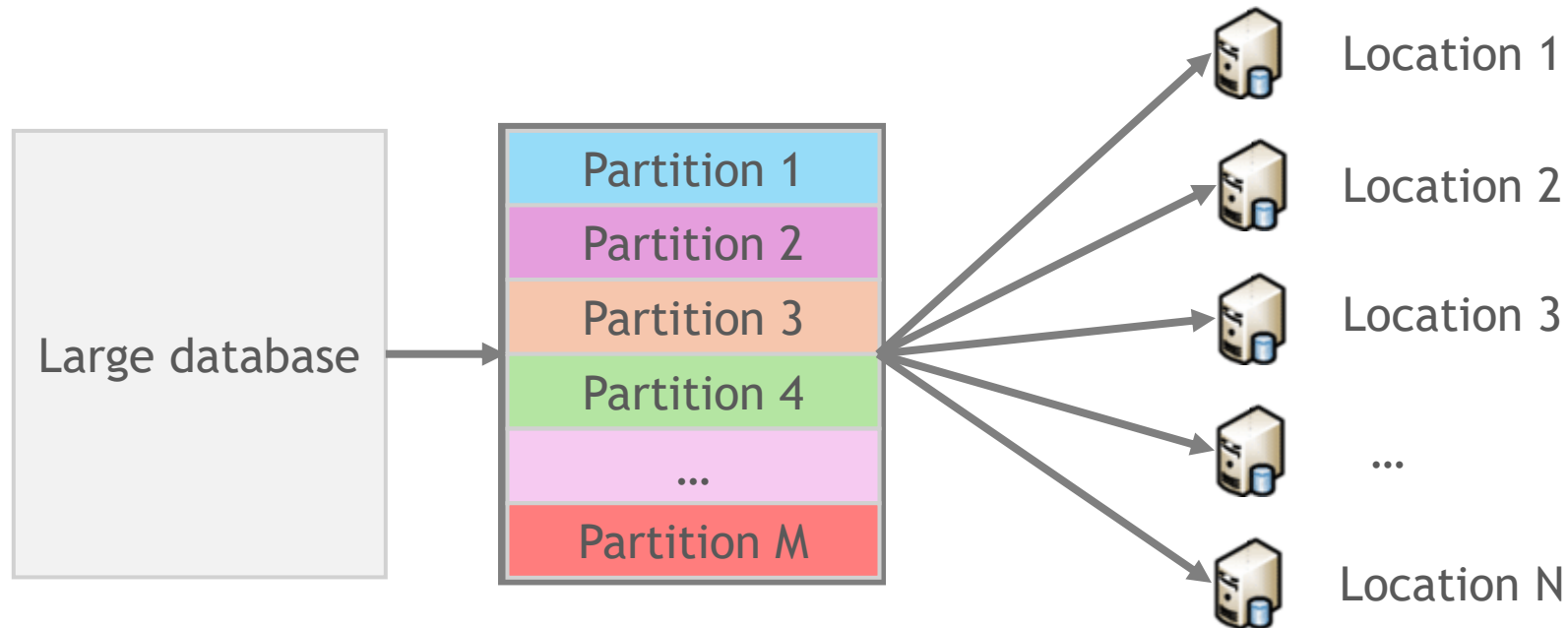
HP2

CUSTOMER ID	FIRST NAME	LAST NAME	FAVORITE COLOR
3	SELDA	BAĞCAN	PURPLE
4	JIM	PEPPER	AUBERGINE

Source: <https://www.digitalocean.com/community/tutorials/understanding-database-sharding>

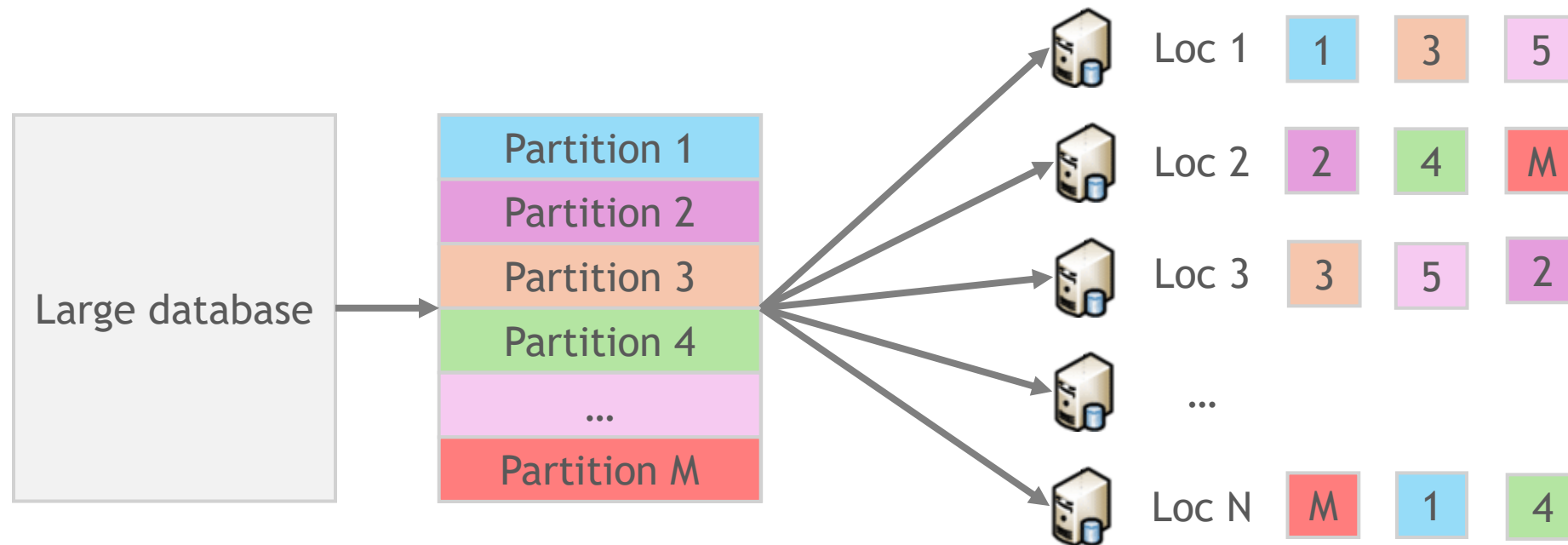
Example of Distribution

- New nodes can be added to handle more load easily (scalability).



Example of Distribution

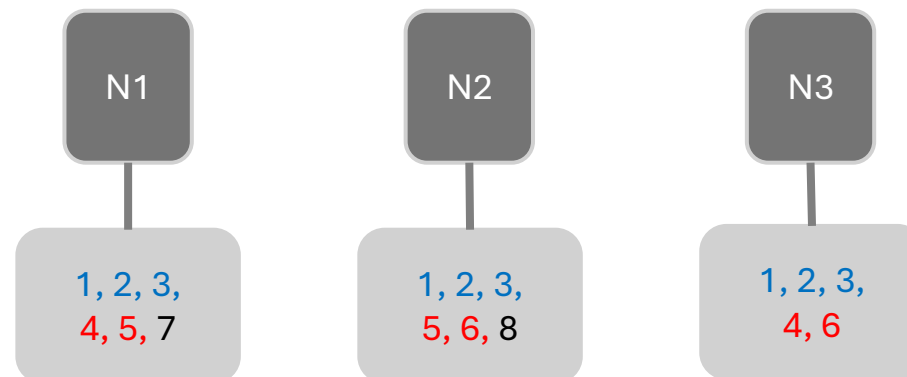
- Replication
 - Example: 2 replicas



Example of Distribution

- Fault tolerant – Node failure

Table name	Replication factor	Partition ID
Table 1	3	[1, 2, 3]
Table 2	2	[4, 5, 6]
Table 3	1	[7, 8]



Distributed Database

- Major concepts

- Concurrency Control – ensures no conflict from multiple users' operations.
- Distributed Query Processing – query optimization must consider where data resides.
- Network failure: trade-offs between *Consistency*, *Availability*, and *Partition Tolerance (CAP Theorem)*.

- Use cases

Company	Distributed DB Used	Reason
Netflix	Cassandra	Global streaming with fault tolerance
Amazon	DynamoDB	E-commerce at global scale
Google	Spanner	Global strong consistency
Facebook	TAO, MyRocks	Social graph management
Uber	Schemaless	Real-time location-based matching

Cloud Database

- A database service that runs on a cloud computing platform.
 - Can be accessed over the internet without needing users to manage physical hardware, installation, or maintenance.
- Database Services
 - Relational (SQL) Cloud Databases: Amazon RDS, Azure SQL Database, Google Cloud SQL
 - NoSQL Cloud Databases: DynamoDB (key-value), Cosmos DB (multi-model), Firebase Realtime Database



Storage



Dedicated Server

One physical machine runs a single app of a business

Expensive & high maintenance



Virtual Private Server

One physical machine runs multiple apps of a business on virtualized machines



Shared Hosting

One physical machine shared by several businesses

Cheap & limited

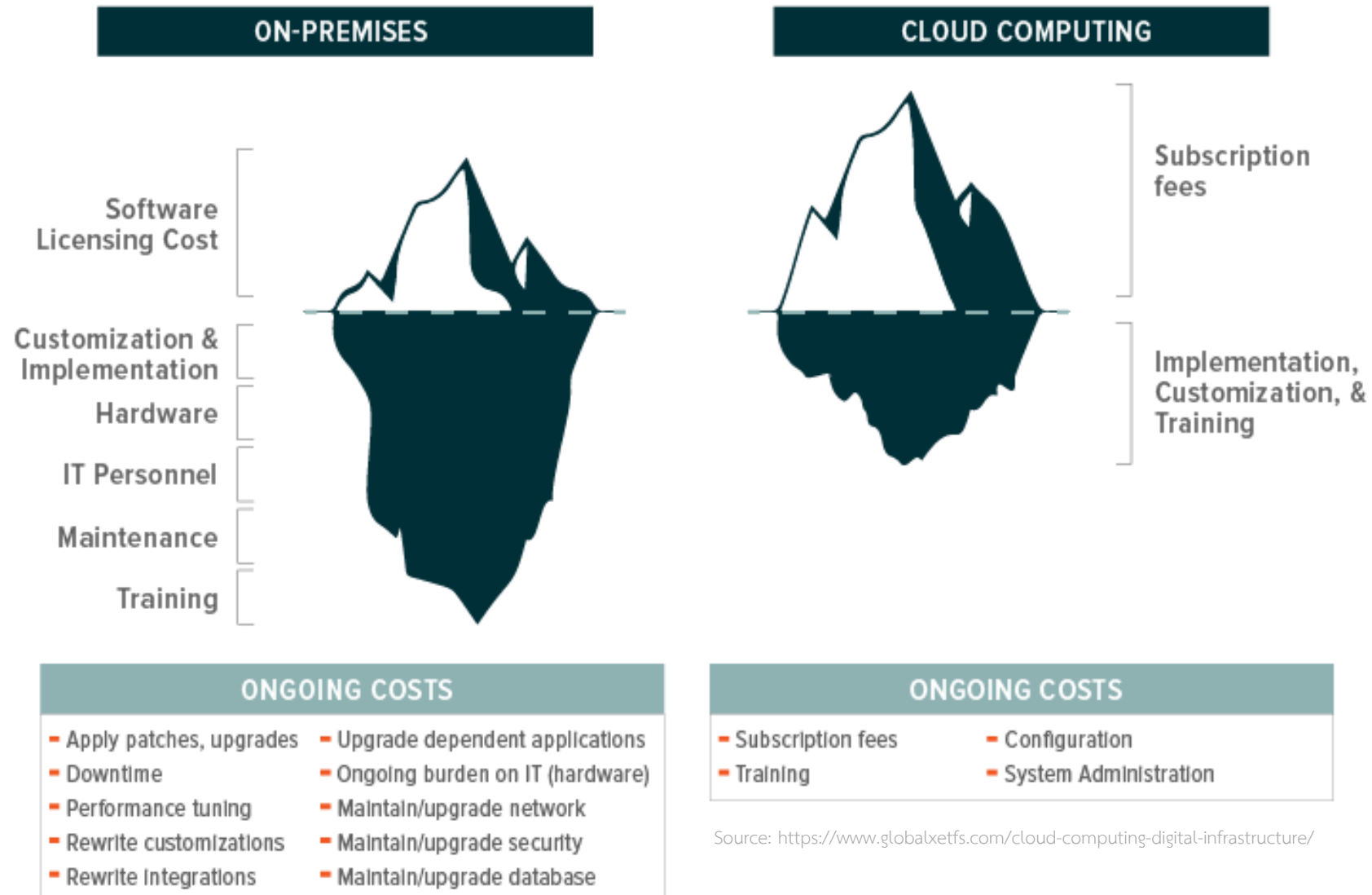


Cloud Hosting

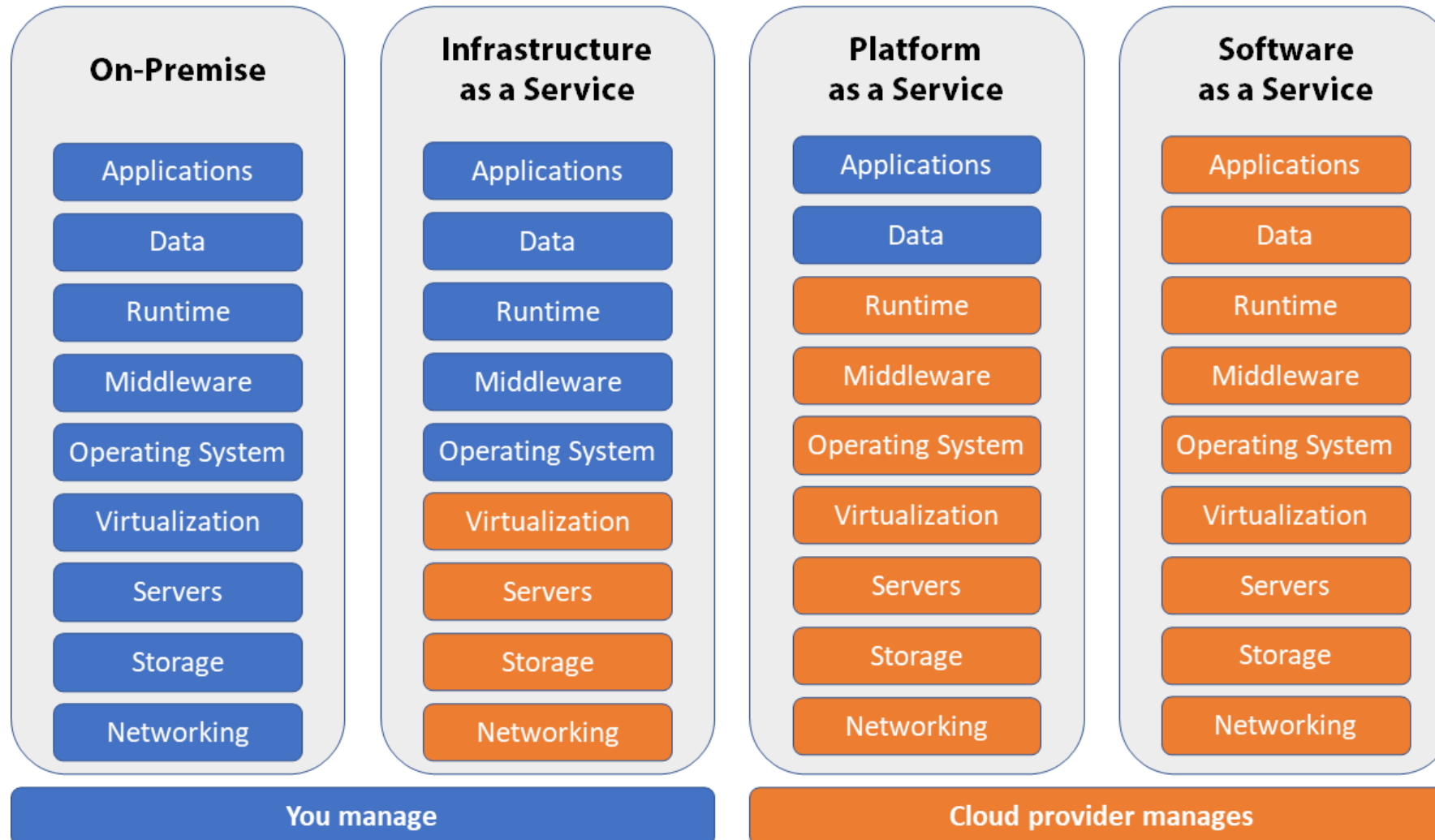
Multiple physical machines (as one system) provide multiple cloud services

Flexible, scalable, secure, cost-effective

On-premise vs Cloud Computing



Types & Responsibilities



Cloud Computing – Key Characteristics

- **Availability** - Ability to **ensure a service** remains available
- **Scalability** - Ability to **grow rapidly** or unimpeded
- **Elasticity** - Ability to **shrink and grow** the capacity based on the demand of traffic, memory, and computing power
- **Fault Tolerance** - Ability to **prevent** a failure
- **Disaster Recovery** - Ability to **recover** from a disaster/failure & prevent the loss of data

Availability

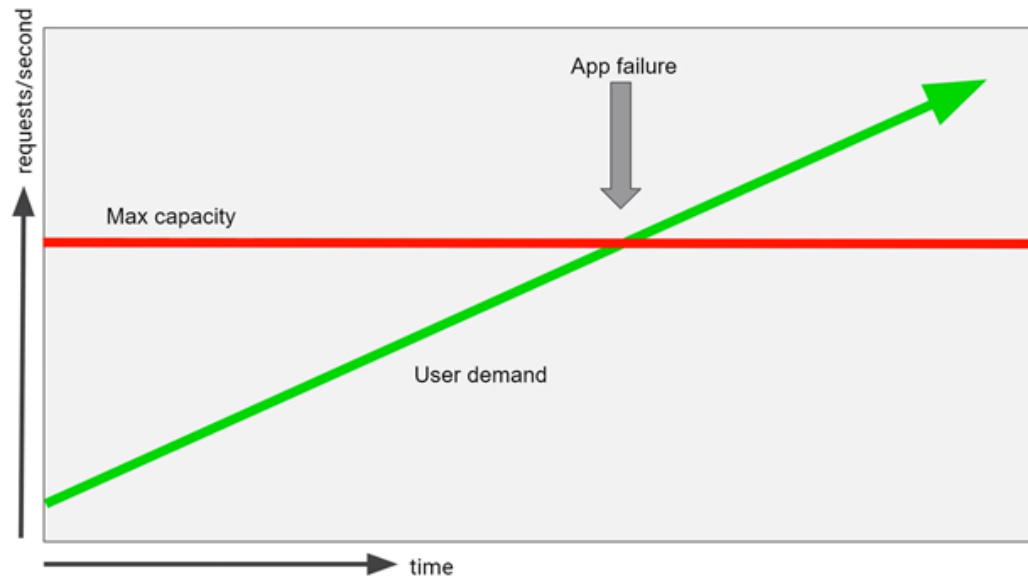
- Ability to ensure a service remains available
- Ensure a certain level of performance - a percentage of time that the service will be responding to requests without problem
 - 90% ~2.4 hours/day
 - 99% ~15 minutes/day
 - 99.9% ~1.45 minutes/day
 - 99.99% ~8.6 seconds/day
 - 100% ???

High Availability (HA)

Said to be impossible

Scalability

- Ability to **grow rapidly** or unimpeded
 - Vertical Scaling – upgrade to a bigger server
 - Horizontal Scaling – add more servers of the same size



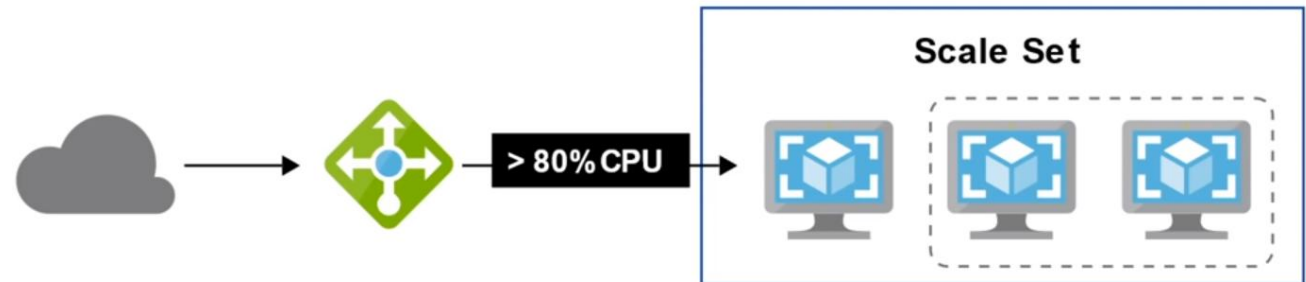
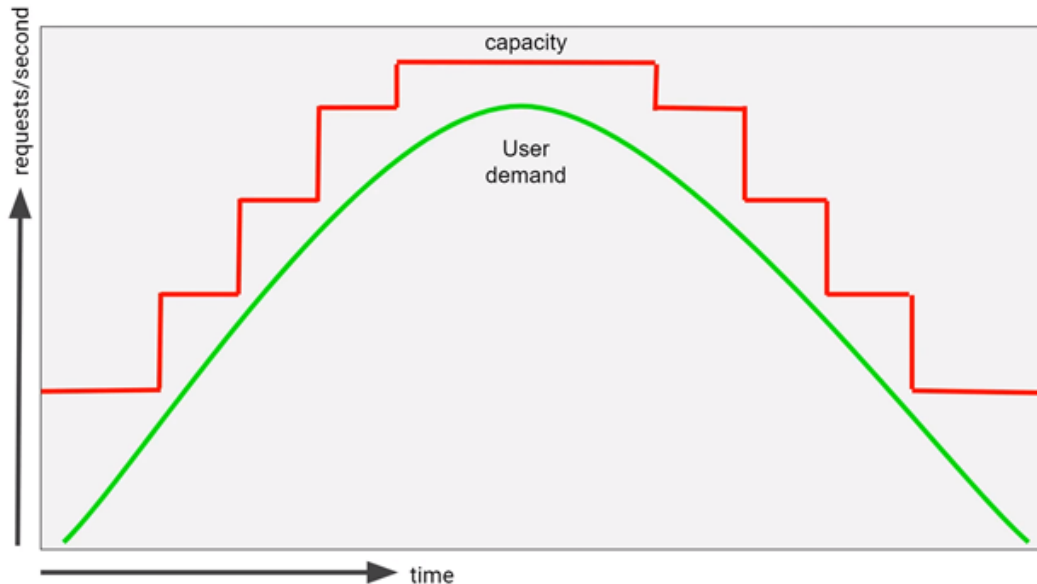
Vertical Scaling



Horizontal Scaling

Elasticity

- Ability to **shrink and grow** the capacity based on the demand of traffic, memory, and computing power
- The system can detect when it's getting busy & add more resources
- Horizontal Scaling: scaling out/in



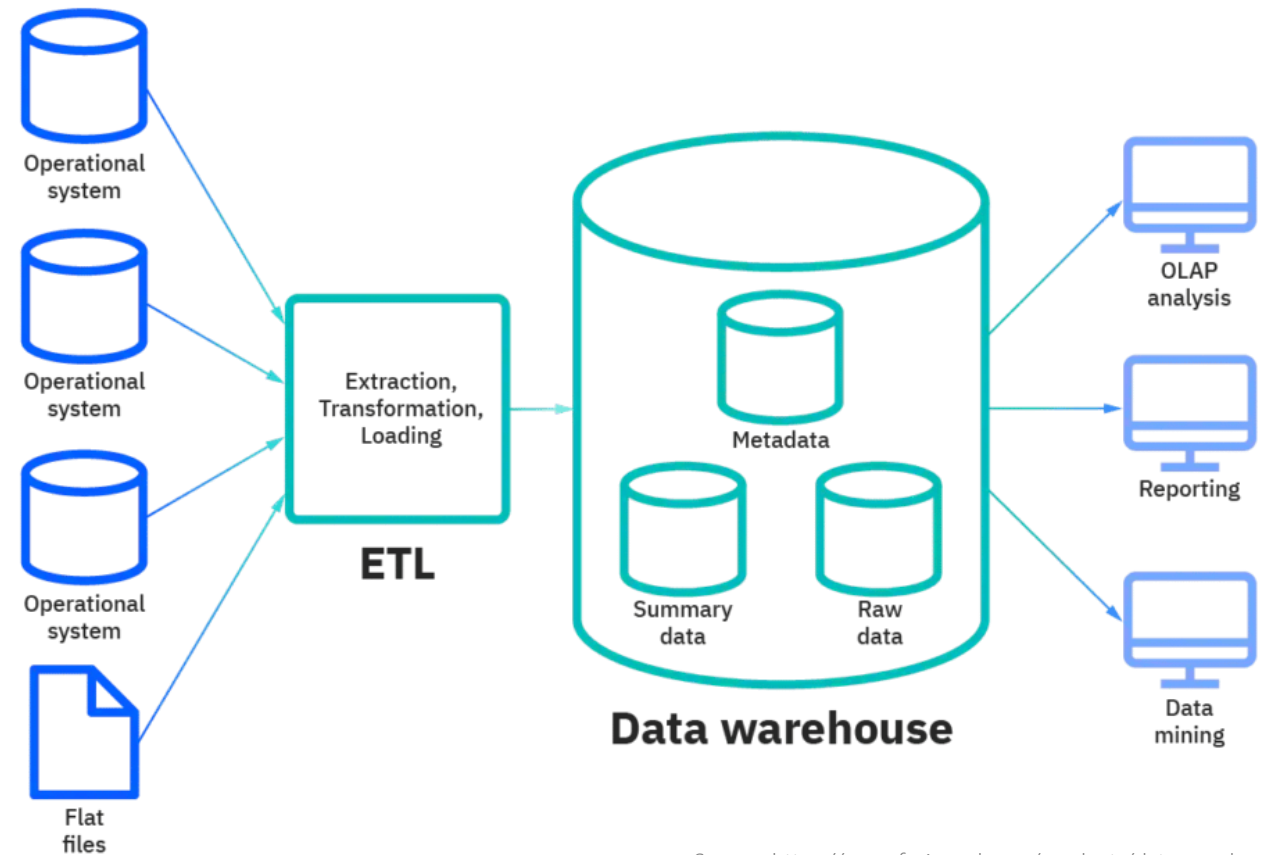
AZ service - VM Scale Sets
Automatically increase/decrease
in response to demand

Data Warehouse

Acts as a centralized, structured **repository designed to store** *processed, cleaned, and organized data* **for analysis and reporting.**

Highlights

- Structured Data Storage with ETL Process
- Schema-on-Write (a predefined schema)
- Optimized for analytical queries
- High Query Performance for complex OLAP queries.
- Stores historical snapshots of business operations for trend analysis.



Source: <https://www.fusionsol.com/products/data-warehouse/>

Data Warehouse Characteristics

Subject-oriented

Delivers information about a **subject** (more defined) instead of organization's current operations.

Sales, distributions, marketing etc.

Integrated

Founding a shared entity to scale the similar data from the different databases (heterogeneous sources).

Must have reliable naming conventions, format and codes.

Data Warehouse Characteristics

Time-Variant

Data is stored with a **time dimension**, allowing for analysis of data over time.

Data is maintained via different intervals of time such as weekly, monthly, or annually etc.

Once data is stored in the data warehouse then it cannot be modified, alter, or updated.

Non-Volatile

The data resided in data warehouse is **permanent**.

Data is read-only and refreshed at particular intervals to **maintain the historical data**.

Frequent changes in operational database is not reflected in the data warehouse.

CRUD functions are lost in data warehouse environment.

Two types of data operations done in the data warehouse are data loading and data access.

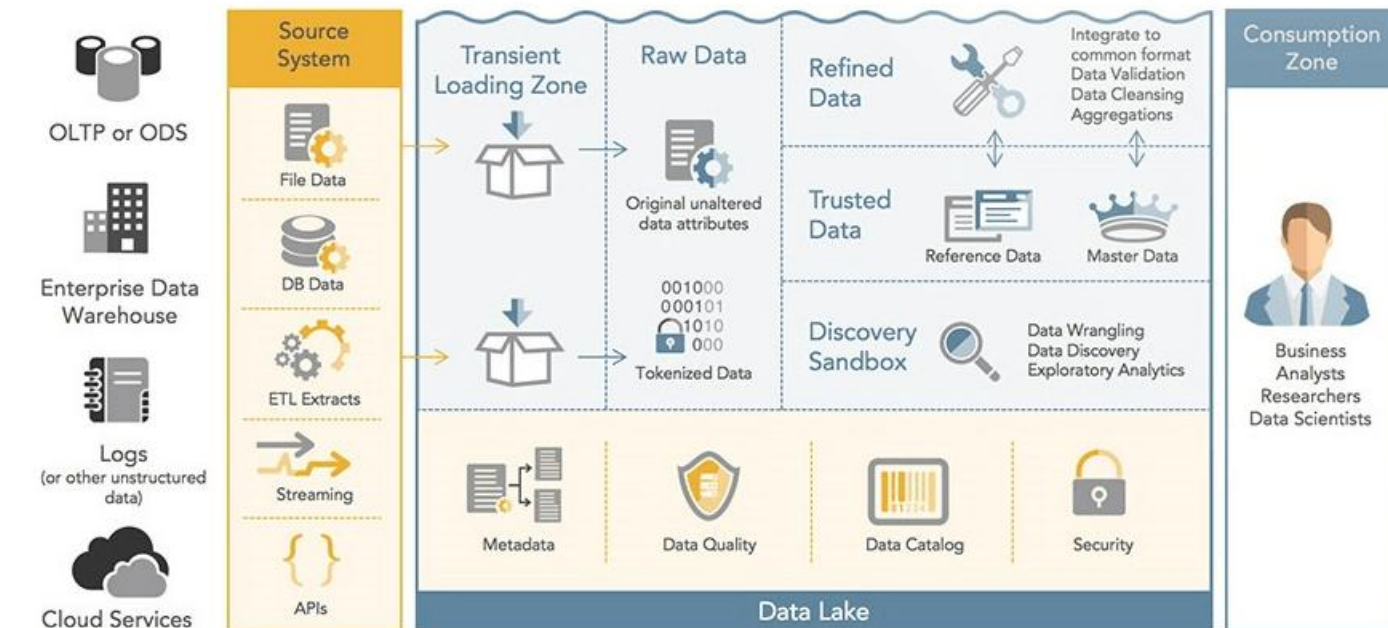
Data Lake

A large storage repository that holds raw, unprocessed data in its **native format until it's needed**.

Highlights

- Structuring Raw Data: data variety and volume.
- Schema-on-Read - schema is applied only when it is read and queried.
- Scalable and Flexible
- Low Cost per Storage Unit - compared to structured, performance-tuned warehouses.
- Supports Advanced Analytics and Machine Learning
- Integration with Data Processing Tools
- Metadata Management

Data lake reference architecture



Source: <https://thanachart.org/2022/01/18/big-data-architecture>

Database, Data Warehouse, Data Lake

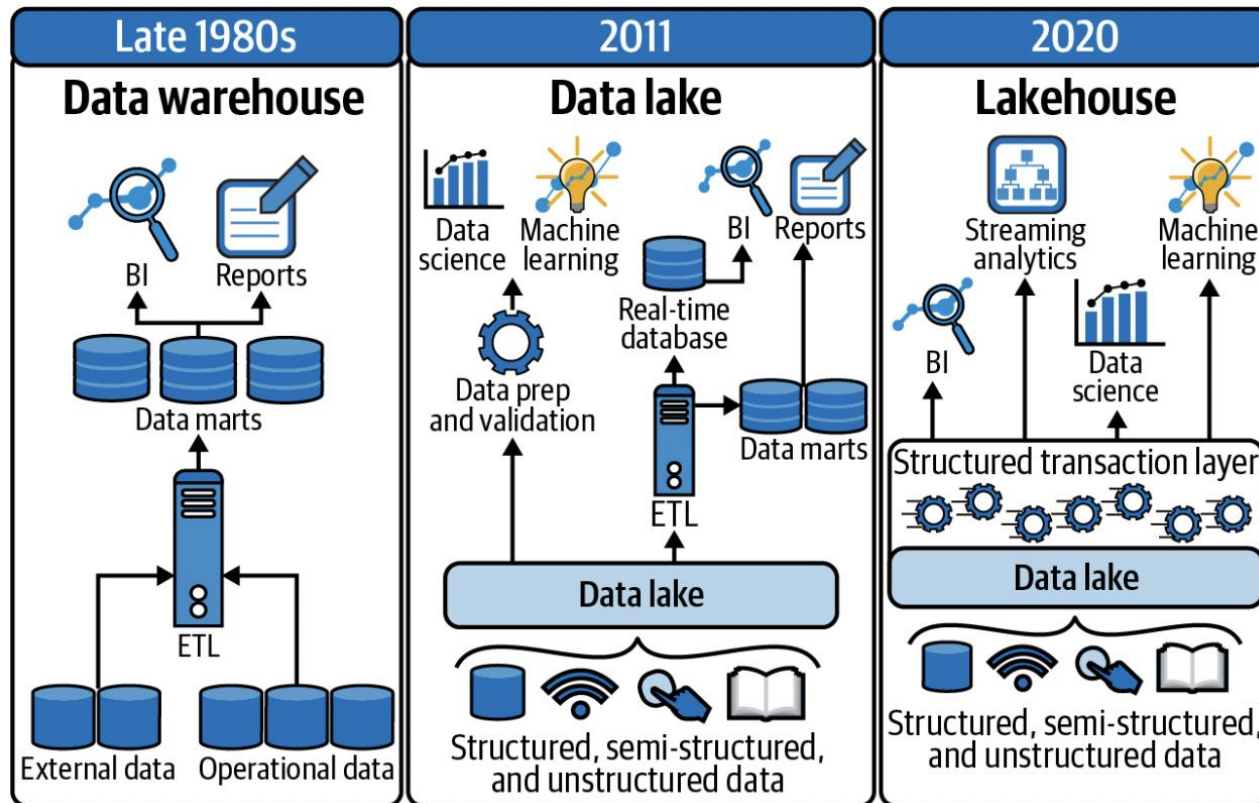
- **Database**
Running the business (real-time data).
- **Data Warehouse**
Analyzing the business over time (clean, structured, historical data).
- **Data Lake**
Exploring possibilities (raw, big, messy data for future use).

Database, Data Warehouse, Data Lake

Aspect	Database	Data Warehouse	Data Lake
Purpose	Store and manage current operational data (transactions, records)	Store historical, cleaned data for analysis	Store raw, diverse data for flexible future use
Data Type	Structured	Structured	Structured, Semi-structured , Unstructured
Data Storage	Normalized and transactional	Preprocessed, aggregated, structured	Raw and unprocessed
Schema	Strict, enforced when writing (schema-on-write)	Strict, enforced when writing (schema-on-write)	Flexible, applied when reading (schema-on-read)
Query Type	Short, fast (CRUD operations)	Complex analytical queries (aggregations, joins)	Custom queries after data exploration
Use Case	Running business operations (OLTP)	Business Intelligence, dashboards, reporting (OLAP)	Big data analytics, AI/ML, archival
Performance Focus	Transaction speed and concurrency	Query speed for large-scale analysis	Storage scalability and flexibility
Cost	Moderate	Higher (optimized for performance)	Lower (optimized for storage)
Example Technologies	MySQL, PostgreSQL, Oracle DB	Snowflake, Redshift, BigQuery	Hadoop HDFS, AWS S3, Azure Data Lake

Data Lakehouse

- To store raw data (like a data lake) while also supporting structured queries, transactions, and governance (like a data warehouse) — all in the same platform



Source: <https://thanachart.org/wp-content/uploads/2022/01/screen-shot-2565-01-21-at-14.08.59.png>